Vegetation Community Mapping of Estero Sargento

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Abstract

Estero Sargento lies at the northern end of the Canal de Infiernillo on the coast of Sonora, Mexico. The estuary is managed by the Comcaac, an indigenous tribe of the region, and is located on their sovereign land. Due to its management by the tribe, the wetland has remained relatively pristine and has seen little development. Generally, the wetlands of the Northern Gulf have received little attention as sites for environmental protection, despite the critical role that they play as stopover sites for migratory birds and as spawning areas for fish and marine invertebrates. Estero Sargento currently has no environmental protection status. In order to aid in ecosystem monitoring we have created a vegetation map using remote sensing with LANDSAT imagery. Spectral classes were predetermined and subsequently ground-truthed using randomized relevé plots. Ground-truthed data was in turn used to re-establish our spectral classes into a working map. Seven primary vegetation communities were defined: mangrove forest, cactus scrub, foredune, backdune scrub, mixed sparse halophytic shrubland, dense halophytic shrubland, and low-growing halophytes. Two vegetation communities remain undefined, as they were too far a distance to reach during the ground-truthing expedition. Further ground-truthing of all defined and undefined communities will be needed in order to create a more detailed and accurate vegetation map of the area. This current working map will be of use in future studies and monitoring of the wetland, and will be added to a small geospatial database managed by Prescott College's Kino Bay Research Center.

Introduction

Estero Sargento is an estuarine wetland along the coast of the Midriff Islands region of the Gulf of California, 15 miles south of Desemboque, Sonora, Mexico. The estuary is located on land belonging to the the Comcaác, a group of people indigenous to the Sonoran coast of the Gulf of California. It is perched at the Northern end of the Canal del Infiernillo, a narrow body of water separating Isla Tiburon from mainland Sonora (Figure 1). Due to the estuary's location on Comcaac land, it has remained relatively pristine and has not been the target of development or overuse. However, the estuary currently has no environmental protection status, despite its importance to both the wildlife and people of the region.

Estuaries along the Gulf are critical resting and refueling sites for migrating birds (Brusca et al. 2006). The degradation of these wetlands can have a profoundly negative affect on species that rely on them as stopover sites through the arid Sonoran Desert. Estero Santa Rosa, further south down the Infiernillo, provides habitat for over 70 species of migratory birds (Chatham et al, 2011). Additionally, these estuaries are important spawning grounds to many species of fish and marine invertebrates (Glenn et al, 2006).

To the Comcaác, Estero Sargento is a site of cultural heritage as well as an important economic resource. The mangroves have long been a place of refuge and abundance among the harsh desert landscapes of the Sonoran Desert. The Comcaác have depended on fishing in coastal regions as a primary means of income since at least the 1920's (Felger and Moser 1985). Estero Sargento continues to be an economic resource for local fishermen through the harvesting of various species of mollusks such as clams, oysters, and scallops, as well as fish such as mullet.

Fig 1 Map showing Isla Tiburón, el Canal de Infernillo, and Estero Sargento (extent outlined in red)

Today, estuaries all over the world are at risk; 35% of worldwide mangrove cover has disappeared, and in some countries up to 80% has been lost (Wells and Ravilious 2006). In Mexico, the estuaries of the Northern Gulf are not well-studied and, consequently, many have not been targeted as areas of conservation. Many of these estuaries have become sites of human development projects such as resorts, marinas, and aquaculture farms (Glenn et al. 2006). This type of development causes great harm to these pristine wetlands. Discharge from aquaculture farms both pollutes and changes the shape of the estuaries themselves (Paez-Osuna et al. 1998). Protecting these wetlands is essential for both the wildlife that live there and people that rely on them.

Students of Prescott College along with professors have been mapping esteros of the Northern-Central Gulf in ongoing projects since 2008 with the aim of assisting conservation efforts (Atteberry et al, 2012, Brown et al, 2010). As a continuation of this project, we have created a vegetation map of Estero Sargento using satellite imaging and GIS software. The working map is intended to serve further research of the Estero and assist in Comcaac ecosystem monitoring efforts.

Methods

For the purposes of efficiency we employed a stratified random sampling technique to sample the floristics of the Estero. This was done in a way that was faithfully representative while requiring the fewest number of plots. To come up with vegetation strata, we chose to forgo the reconnaissance and entitation techniques described by Mueller-Dombois and Ellenberg (1974). This is because we were unable to visit the site prior to field data collection. Instead, we chose to define the strata using remote sensing and GIS analysis.

Spectral Classification

For the sake of objectivity, we employed an unsupervised classification technique (Eastman 1997). This technique employs an algorithm that analyzes satellite imagery for unique reflectance patterns, on the assumption that such patterns represent distinct land cover classes. We used LANDSAT imagery of the Estero taken on 19 January, 2019, courtesy of USGS. Pre-processing of this data involved generating a transformed soil-adjusted vegetation index (TSAVI) developed by Baret et al. 1989, as described in Thiam and Eastman (1997). This vegetation index was optimal because it corrects for soil reflectance in ecosystems with relatively high bare soil, as is the case in the Sonoran desert.

We created a data processing mask that encompassed only the area of the estuary itself. The TSAVI was input into ArcGIS software along with LANDSAT imagery bands 3 and 6 (green and nearinfrared 1) to perform an isocluster unsupervised classification (see Figure 2). At our designation, the program produced six unique spectral classes. Through trial and error, we determined that designating more than six classes caused the algorithm to split up vegetation communities too much, resulting in multiple classes representing the same community.

Field Sampling

Once the strata were defined through the generation of spectral classes, we placed 5 sample plots in each class. We chose the location of plots within each class while considering both accessibility and the proximity of sample sites to one another. These locations were manually digitized on-screen and the coordinates were uploaded to the hand-held GPS units.

We then travelled to the estero to ground-truth the connection between the computer-generated classifications and the actual floristic classes, using the relevé technique. One group used a Trimble Juno GPS unit to navigate to seven of the predefined locations, while another group sampled at random locations using the subjective entitation technique described by Mueller-Dombois and Ellenberg (1974).

For each sample plot, we performed the relevé technique for vegetation community sampling (Mueller-Dombois 1974). At each location, a ten-by-ten plot was outlined. Observers then compile a full floristic list of species present in the plot. For each species, observers assign a score from the Braun-Blanquet (1965) cover-abundance scale (see Table 1). A GPS point is taken at the southeast corner of the plot so that it can later be related to the unsupervised classification spectral class. In total, we sampled 21 relevés.

Floristic Classification

A multivariate analysis (TWINSPAN) was performed on the relevé data to parse out different vegetation communities based on the abundance and uniqueness of the species within them. The software we used was PCOrd (McCune and Mefford 2016). A dendrogram of the resulting division of communities can be found in Appendix A. We used the result of this analysis along with the unsupervised classification map to make informed decisions on how to develop the final vegetation map.

Braun-Blanquet Score	*Range of Cover*
5	75 - 100 %
4	50 - 75 %
3	25 - 50 %
2	5 - 25 %
1	<5 %; numerous individuals
+	<5 %; few individuals
r	very few individuals

Table 1 The Braun-Blanquet cover-abundance scale. Scores *5, 4, 3,* and *2* are estimates of cover for shrub- and tree-level species, whereas *1, +,* and *r* are estimates of abundance for smaller plant life forms.

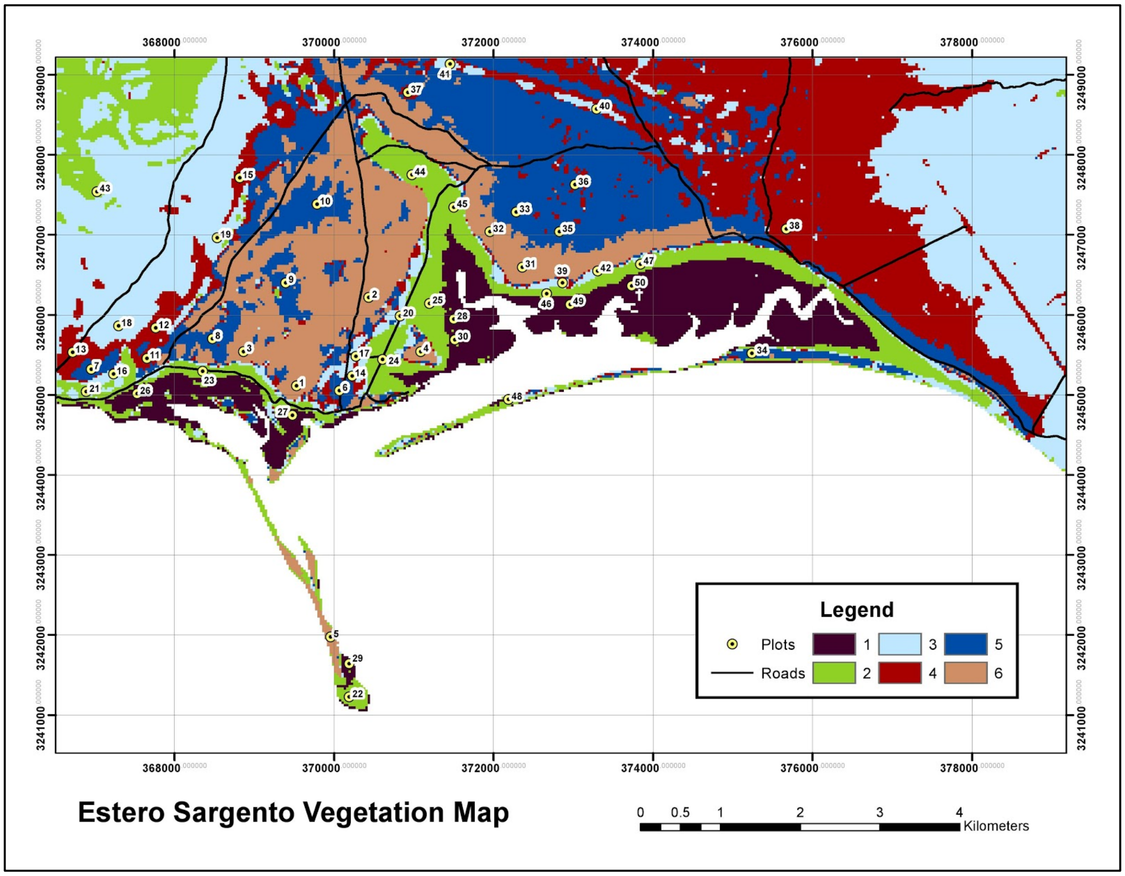

Fig 2 Spectral classes generated by an isocluster unsupervised algorithm. This map was used to designate relevé locations, and was our working map used in the field.

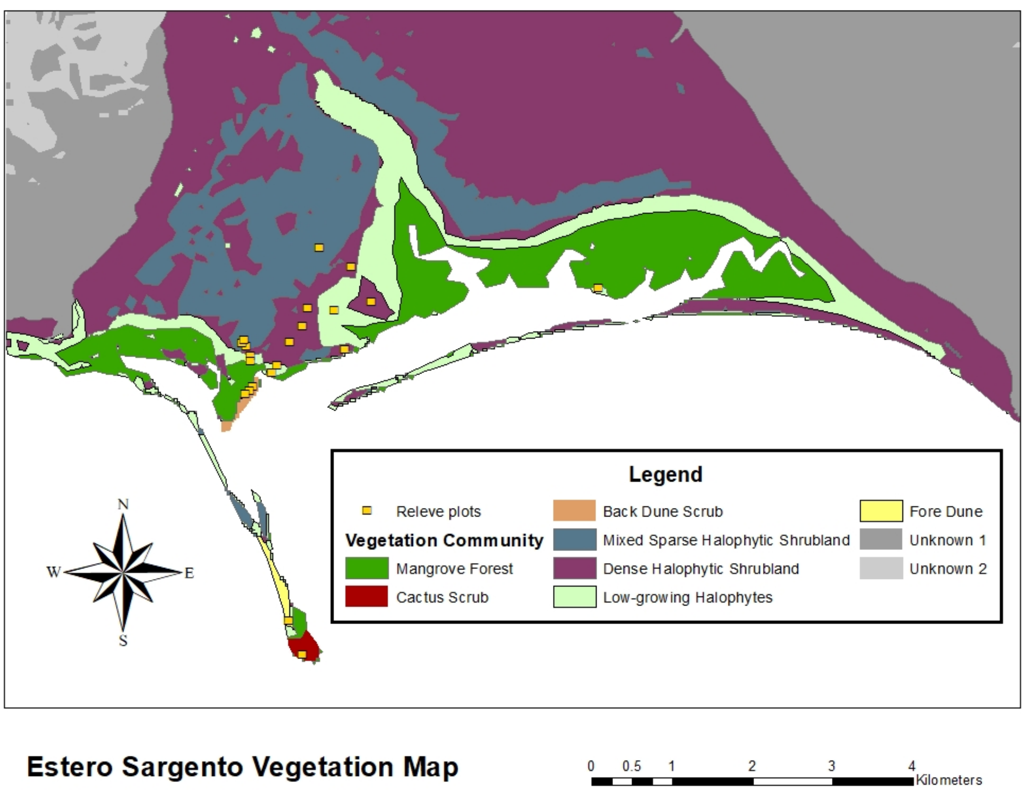

Fig 3 The final vegetation map of Estero Sargento.

Results

We classified seven different vegetation communities in our final map (Figure 3). We also included two unknown spectral categories representing the areas outside of our sampling range, which could be clarified and described with future sampling. The name of each community is listed below, followed by a general description of the vegetation and the names of key species.

Mangrove Forest

This dense, virtually impenetrable community is dominated by *Avicennia germinans* and *Rhizophora mangle*, with some scattered *Laguncularia racemosa*. The mangrove forest community lines the canals along the interior of Estero Sargento and is characterized by halophytic trees whose trunks may be periodically submerged by tidal fluctuations. The understory includes species such as *Batis maritima, Sesuvium portulacastrum,* and *Distichlis* spp.

Cactus Scrub

This vegetation community is located on steep, rocky terrain and is characterized by tall shrubs and columnar cacti. Dominant species include *Stenocereus gummosus, Pachycereus pringlei, Jatropha cuneata, Encelia farinosa, Lycium andersonii,* and *Cylindropuntia fulgida*, with an understory of *Mentzelia adhaerens, Atriplex barclayana,* and *Mammillaria* spp.

Fore Dune

The Fore Dune community is situated on soft, sandy dunes along the gulf-facing shoreline of Estero Sargento. It is quite sparse relative to most other communities within the estero. The two most dominant plants in the Fore Dune community are *Abronia maritima* and *Sesuvium portulacastrum*, which are both sprawling, halophytic succulents.

Back Dune Scrub

The Back Dune Scrub community is found behind the Fore Dune community and is much more diversely vegetated. Scattered shrubs and dense patches of both tall and appressed forbs characterize this community. The shrub species include *Jatropha cinerea, Frankenia palmeri,* and *Atriplex barclayana.* The forbs of this community are *Cryptantha angustifolia, Croton californicus, Dalea mollis,* and *Palafoxia arida.*

Mixed Sparse Halophytic Shrubland

Found on the salt flats behind the mangroves, this halophytic community is characterized by dispersed large shrubs of *Allenrolfea occidentalis* and *Jatropha cuneata* with bare soil and a diverse selection of small appressed forbs. The forbs and smaller shrubs in this community are *Suaeda nigra, Perityle emoryi, Nama demissum, Atriplex linearis,* and *Euphorbia eriantha*.

Dense Halophytic Shrubland

The Dense Halophytic Shrubland is similar to the Mixed Sparse Halophytic Shrubland but with very little bare soil and fewer species. Primarily woody halophytes and salt-tolerant succulents, this vegetative community completely covers the ground, forming a thick mat. *Allenrolfea occidentalis, Frankenia palmeri, Salicornia bigelovii, Sesuvium portulacastrum,* and *Monanthochloe littoralis* are the dominant species from this community.

Low-growing Halophytes

The Low-growing Halophyte community fringes the mangroves and is comprised of short shrubs, grasses, and salt-tolerant succulents. The appearance of this community is somewhat similar to the Dense Halophytic Shrubland; however, the larger shrubs are absent and there is a greater diversity of succulents. The species found in this vegetative community include *Suaeda esteroa, Atriplex barclayana, Distichlis palmeri, Monanthochloe littoralis, Salicornia bigelovii,* and *Sesuvium portulacastrum.*

Unknown 1 and Unknown 2

We were unable to sample these areas, but our isocluster unsupervised classification divided the areas into two different spectral classes. Based on visual entitation and a topographic map of these areas, the vegetation communities are most likely similar to the Cactus Scrub community described above. Unknown 2 appears to be a steeper, more sparsely populated community than Unknown 1.

Discussion

Coastal wetlands such as Estero Sargento play a key role in the overall health of coastal zones as they provide essential habitat for numerous flora and fauna. Estero Sargento, specifically, is a location of great cultural and economic significance to the Comcaác people indigenous to the area. Through fishing and harvesting of aquatic invertebrates, the Comcaác people have found economic value in the estuary. This economic value, along with its cultural value, is a key factor behind the need to protect Estero Sargento from further environmental deterioration. There is a lack of geospatial information concerning the vegetation of Estero Sargento, which may cause obstacles to future conservation efforts. This project provides preliminary geospatial data in the form of an ArcGIS geodatabase, as well as a map (Figure 3) which can be interpreted more broadly by a wider audience. These materials will contribute to a larger collection of similar vegetation maps of estuaries in the Northern Gulf of California, which may aid in conservation efforts of these critical coastal wetland habitats.

We utilized remote sensed imagery along with spatial analysis software to come up with our initial classification map. Using this map, we designated ground-truth points and sampled the actual flora on the ground within each class by using the relevé technique. Using a TWINSPAN analysis, we generated a dendrogram that split up these sample plots into five distinct vegetation communities (Appendix A). The TWINSPAN analysis, along with data collected from the field, aided in a manual reclassification of some of the vegetation communities. The newly classified communities included Dense Halophytic Shrubland and Mixed Sparse Halophytic Shrubland, because the ArcGIS isocluster classification seemed to be influenced more by plant population density than by actual community distinction. In the field, we were surprised to find the Cactus Scrub community located on the southernmost tip of a peninsula extending out from Estero Sargento. This area was the only area in Estero Sargento consisting of Cactus Scrub, even though our initial field maps did not reflect its presence in the estuary.

Our final vegetation map provides an informed delineation of vegetation communities within Estero Sargento, although accuracy and precision can certainly be improved through future work. While in the field, we noticed a number of species that were present in certain communities, but were not captured by our sample plots. This was the case in the Back Dune community, where we noticed several unique species just outside the boundary of the plots. This could be because we did not perform the nested-plot method to determine our relevé plot size; instead we chose to re-use the plot size that previous classes had used in other estuaries. Distance also proved to be a limitation, as we did not sample the eastern half of the estero at all.

Our map is a preliminary iteration of a project that should continue to be improved upon. In order to further advance this project, more sample plots must be studied, specifically in eastern regions of the estero. Further sampling within the communities would also allow improvement in the overall accuracy of the field map and the vegetation classes outlined within it.

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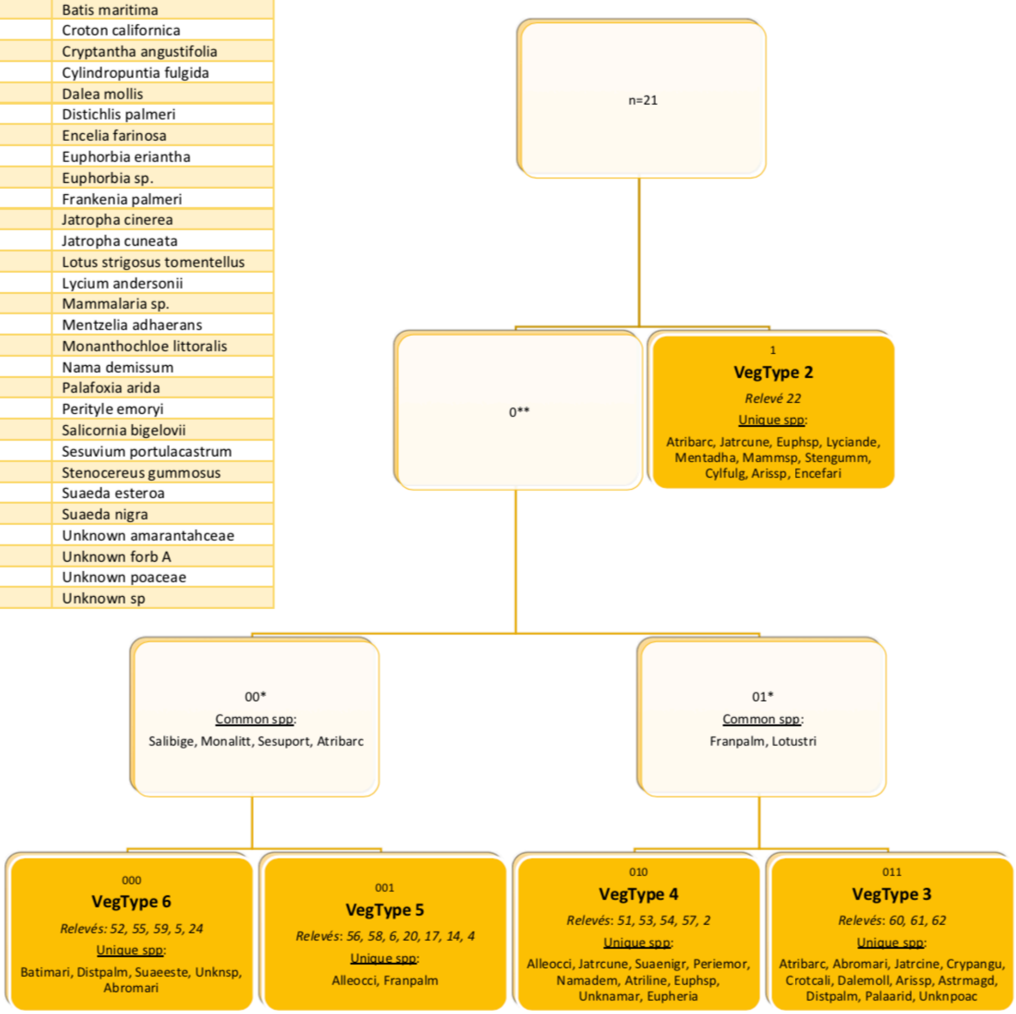
Appendix A: Relevé Dendrogram

Encifari	Encelia farinosa
Abromari	Abronia maritima
Alleocci	Allenrolfea occidentalis
Arissp	Aristida sp.
Astrmagd	Astragalus magdalenae var. magdalenae
Atribarc	Atriplex barclayana
Atriline	Atriplex linearis
Batimari	Batis maritima
Crotcali	Croton californica
Crypangu	Cryptantha angustifolia
Cylifulg	Cylindropuntia fulgida
Dalemoll	Dalea mollis
Distpalm	Distichlis palmeri
Encefari	Encelia farinosa
Eupheria	Euphorbia eriantha
Euphsp	Euphorbia sp.
Franpalm	Frankenia palmeri
Jatrcine	Jatropha cinerea
Jatrcune	Jatropha cuneata
Lotustri	Lotus strigosus tomentellus
Lyciande	Lycium andersonii
Mammsp	Mammalaria sp.
Mentadha	Mentzelia adhaerans
Monalitt	Monanthochloe littoralis
Namademi	Nama demissum
Palaarid	Palafoxia arida
Periemor	Perityle emoryi
Salibige	Salicornia bigelovii
Sesuport	Sesuvium portulacastrum
Stengumm	Stenocereus gummosus
Suaeeste	Suaeda esteroa
Suaenigr	Suaeda nigra
Unknamar	Unknown amarantahceae
UnknforA	Unknown forb A
Unknpoac	Unknown poaceae
Unknsp	Unknown sp



Appendix B: Estero Sargento Sampled Flora

Aizoaceae
Sesuvium portulacastrum

Amaranthaceae
Allenrolfea occidentalis
Atriplex barclayana
Atriplex linearis
Salicornia bigelovii
Suaeda esteroa
Suaeda nigra

Asteraceae
Encelia farinosa
Palafoxia arida
Perityle emoryi

Bataceae
Batis maritima

Boraginaceae
Cryptantha angustifolia
Nama demissum

Cactaceae
Cylindropuntia fulgida
Mammillaria sp.

Euphorbiaceae
Croton californicus
Euphorbia eriantha
Euphorbia sp.
Jatropha cinerea
Jatropha cuneata

Fabaceae
Astragalus magdalenae var. magdalenae
Dalea mollis
Lotus strigosus var. tomentellus

Frankeniaceae
Frankenia palmeri

Loasaceae
Mentzelia adhaerens

Nyctaginaceae
Abronia maritima

Poaceae
Aristida sp.
Distichlis palmeri
Monanthochloe littoralis

Solanaceae
Lycium andersonii